

KATHMANDU UNIVERSITY
End Semester Examination
August/September, 2016

Mark scored :

Level : B. Sc.

Year : I

Exam Roll No. :

Time: 30 mins.

Course : MATH 101

Semester : I

F. M. : 20

Registration No.:

Date :

SECTION "A"
[10 Q. $\times$ 1 = 10 marks]

Fill in the blanks space(s) by most appropriate word(s) or symbol(s).

1. The domain of the function $f(x) = \sqrt{4-x}$ is _____
2. The vectors $\vec{e}_1 = (1, 0)$ and $\vec{e}_2 = (0, 1)$ in R^2 are _____
3. The value of $\frac{d}{dx}(\log(\tan x))$ _____
4. The inclination of the tangent to the curve $x^2 + y^2 = 36$ with x -axis at $(2, 6)$ is given by _____
5. The row rank of a matrix is the number of _____ in row echelon form.
6. A function $y = f(x)$ is even function if _____
7. The function $y = f(x)$ has a jump discontinuity at $x = a$ if it has a condition _____
8. The process idea of Gaussian elimination is to introduce zeros in the augmented coefficient matrix by means of _____
9. $\int_0^2 |1-x| dx =$ _____
10. $\lim_{x \rightarrow -2^+} \frac{a^x - b^x}{x} =$ _____

SECTION "B"
[10 Q. $\times$ 1 = 10 marks]

11. The series $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converges for all values of x with _____
 $[-1 \leq x \leq 1; \quad -1 < x \leq 1; \quad -1 \leq x < 1; \quad -1 < x < 1]$
12. A function will have a stationary point at the point when _____
 $[f''(x) > 0; \quad f''(x) = 0; \quad f'(x) > 0; \quad f'(x) = 0]$
13. The average value of $f(x) = \frac{1}{x}$ on the closed interval $[1, 3]$ is _____
 $[1/2; \quad 2/3; \quad \ln 2/2; \quad \ln 3/2]$
14. The value of $g(x) = x^2 + 6x + 10$ is minimum at _____
 $[x = -10; \quad x=1; \quad x = -3; \quad x = -1]$

15. A function f is said to have _____ value at $x = c$ if there exist an open interval (a, b) containing the point $x = c$ such that $f(c) \geq f(x), \forall x \in (a, b)$.
 [absolute maximum; relative maximum;
 absolute minimum; relative minimum]
16. The graph of the function $y = x^2 + 2$ is symmetric about _____.
 [y-axis; x-axis; line $y = x$; origin]
17. The oblique asymptote to the curve $y = \frac{x^2-3}{2x-4}$ is _____.
 [$y = \frac{x}{2} + 1$; $y = \frac{x}{2} - 1$; $y = -\frac{x}{2} + 1$; $y = \frac{x}{2} + 2$]
18. The value of $\frac{d^n}{dx^n}(e^{nx})$ is -----
 [$n^n e^{nx}$; $n! e^{nx}$; ne^{nx} ; $n^n e^n$]
19. The mean value theorem holds for the function $y = x^2$ on $[0, 2]$ when c is _____.
 [1; 2; 1.5; 0.5]
20. If F and f are continuous function such that $F'(x) = f(x)$ for all x , then $\int_a^b f(x)dx$ is _____.
 [$F(b) - F(a)$; $F'(b) - F'(a)$; $f'(b) - f'(a)$; $f(b) - f(a)$]

KATHMANDU UNIVERSITY
End Semester Examination
August/September, 2016

Level : B. Sc.
Year : I
Time : 2 hrs. 30 mins.

15 AUG 2016

Course : MATH 101
Semester : I
F. M. : 55

SECTION "C"

[3Q. × 7 = 21 marks]

1. Define the solution of a multisystem. Also, solve the following two system simultaneously and verify your result: [1+4+2]

$$\begin{array}{rcl} x - 2y + 3z = 2 & & x - 2y + 3z = -2 \\ 4x - 9y + 6z = 2 & & 4x - 9y + 6z = 1 \\ -2x + 3y - 5z = -3 & \text{and} & -2x + 3y - 5z = -1 \end{array}$$

OR

Define the row rank of a matrix. Also, find the rank of coefficient and augmented matrix and then check the consistency of the given system of equations: [1+2+4]

$$\begin{array}{rcl} x + 2y - 3z + w = 2 \\ -3x - 8y + 4z - 4w = -5 \\ -2x - 4y + 5z - 5w = -7 \\ 2x + 4y - 4z - 2w = -10 \end{array}$$

2. Define absolute extremum values of the function $y = f(x)$ defined on a closed interval. If $f(x) = -x^3 + 12x + 5$, $-3 \leq x \leq 3$, then investigate rise, fall, and extreme value(s) of it. [2+1+1+3]

3. Write the condition for the existence of the limit of a function at a point. Prove that if $f(x)$ is differential at $x = c$, then $f(x)$ is continuous at $x = c$. Also, show that

$$f(x) = \frac{x^2 - 16}{x^2 - 3x - 4} \text{ has a continuous extension at } x = 4, \text{ and find its extension.}$$

[1+3+3]

SECTION "D"

[6Q. × 4 = 24 marks]

4. Find $\frac{dy}{dx}$ (ANY TWO):

(a) $y = xe^x$

(b) $x^2 - 2xy + y^2 = 4$

(c) $x = e^{\log \cos 4\theta}$, $y = e^{\log \sin 4\theta}$

5. Evaluate the following integrals (ANY TWO):

(a) $\int_0^{\pi/2} \frac{dx}{1 - \sin x}$

(b) $\int \frac{dt}{t^3 + t^2 - 2t}$

(c) $\int_0^\infty \frac{\ln x}{x^2} dx$

6. Define vertical asymptote of the graph of $y = f(x)$. Also, find the asymptote(s) to the graph of $y = \frac{2x^2-3}{7x+4}$.
7. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the line $y = 1, x = 4$ about the line $y = 1$.

OR

Evaluate the integral $\int_0^1 (x^2 + 1) dx$, using the limit of sum with inscribed rectangle method.

8. Define basis of a vector space. Also, find a basis for the subspace S of R^3 that is the plane with equation $4x - 2y + 7z = 0$
9. Verify the convergence or divergence of the series $\sum_{n=1}^{\infty} \frac{(n+3)!}{3!n!3^n}$.

SECTION "E"[5Q. $\times$ 2 = 10 marks]

10. Show that $T: R^2 \rightarrow R^3$ defined by $T(x, y) = (x, x + y, 2y)$ is linear mapping.
11. Solve the initial value problems: $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$; $y(0) = 0$
12. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$.
13. Find the average value of the function $y = \sin x$ on $[0, \frac{\pi}{2}]$.
14. Show that $\vec{x} = (2, 3)$ is the eigenvector of the matrix $B = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$.